

HANKO RUSSARO, FinlandWMO: **029820**Lat: **59.774N**Lon: **22.952E**Elev: **21**StdP: **14.68**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.3	9.5	0.1	5.5	5.1	5.1	7.2	10.2	44.6	27.9	40.7	30.9	15.7	50	0.803

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	74.4	67.7	72.0	65.8	69.7	64.4	69.0	73.1	67.1	70.9	65.2	68.9	13.4	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.4	100.7	71.7	65.5	94.4	70.0	63.5	87.9	67.9	33.1	73.1	31.6	71.2	30.1	68.9	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 40.6	36.5	32.9	DB	2.4	77.4	7.2	4.1	-2.8	80.4	-7.0	82.8	-11.1	85.1	-16.3	88.1	(4)
(5)			WB	1.8	69.5	7.1	2.9	-3.3	71.6	-7.4	73.3	-11.4	74.9	-16.5	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	44.2	28.6	26.9	30.6	37.9	47.2	56.2	63.6	63.2	56.1	46.5	38.9	33.4	(6)
(7)		DBStd	14.02	8.09	8.85	5.97	4.04	5.11	4.37	4.57	4.01	4.45	5.69	6.24	7.27	(7)
(8)		HDD50	3383	662	647	601	362	118	3	0	0	7	135	334	514	(8)
(9)		HDD65	7665	1127	1067	1066	812	553	266	84	83	269	575	784	979	(9)
(10)		CDD50	1263	0	0	0	0	30	189	420	410	188	26	0	0	(10)
(11)		CDD65	68	0	0	0	0	0	1	40	27	0	0	0	0	(11)
(12)		CDH74	88	0	0	0	0	0	1	70	17	0	0	0	0	(12)
(13)		CDH80	3	0	0	0	0	0	0	3	0	0	0	0	0	(13)
(14)	Wind	WSAvg	16.5	18.6	17.8	16.7	14.8	14.1	14.2	13.4	14.1	16.3	18.3	19.6	20.6	(14)
(15)	Precipitation	PrecAvg	25.9	2.4	1.7	1.6	1.4	1.1	1.6	2.2	3.0	2.2	3.0	3.1	2.6	(15)
(16)		PrecMax	30.0	4.3	3.9	3.7	2.5	2.4	3.0	6.5	6.9	5.7	7.0	6.4	4.7	(16)
(17)		PrecMin	18.8	0.3	0.1	0.2	0.4	0.2	0.6	0.0	0.3	0.6	1.2	0.3	0.7	(17)
(18)		PrecStd	3.2	1.1	1.0	0.7	0.6	0.6	0.7	1.7	1.6	1.3	1.4	1.5	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.1	39.4	42.9	52.8	66.4	71.4	79.1	76.1	67.8	58.4	50.5	46.5	(19)
(20)			MCWB	40.9	37.6	38.6	45.7	56.7	62.0	69.3	68.3	61.9	55.4	49.0	44.9	(20)
(21)		2%	DB	40.4	37.8	40.2	48.6	61.1	67.4	75.9	73.4	65.1	56.7	49.4	44.9	(21)
(22)			MCWB	39.4	36.4	37.2	43.3	54.1	60.0	68.7	66.8	60.5	54.4	48.0	43.4	(22)
(23)		5%	DB	39.0	36.9	38.3	45.8	57.5	65.1	73.4	71.3	63.5	55.3	48.1	43.2	(23)
(24)			MCWB	37.8	35.7	36.1	41.5	51.4	59.0	66.9	65.4	59.5	53.4	46.6	41.6	(24)
(25)		10%	DB	37.5	36.0	36.9	43.6	54.8	62.9	71.0	69.3	61.9	53.7	46.6	41.6	(25)
(26)			MCWB	36.4	35.0	35.1	40.1	49.7	57.9	65.6	64.3	58.5	51.9	44.9	40.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.4	38.4	39.4	46.6	58.1	63.9	72.1	70.6	63.6	56.2	49.8	45.6	(27)
(28)			MCDWB	41.8	38.8	41.8	51.1	64.4	68.3	76.0	74.4	66.4	57.4	50.2	46.1	(28)
(29)		2%	WB	39.6	36.9	37.7	44.1	55.2	61.7	70.0	68.4	61.8	54.9	48.3	43.8	(29)
(30)			MCDWB	40.2	37.4	39.4	47.9	59.9	65.8	74.4	71.9	64.1	56.1	49.2	44.8	(30)
(31)		5%	WB	38.1	36.0	36.6	42.2	52.6	60.0	68.2	66.6	60.3	53.7	46.9	42.0	(31)
(32)			MCDWB	38.9	36.7	37.9	45.1	56.5	63.9	72.4	69.9	62.5	55.1	48.0	43.1	(32)
(33)		10%	WB	36.6	35.2	35.6	40.7	50.3	58.5	66.4	65.1	59.2	52.3	45.2	40.4	(33)
(34)			MCDWB	37.4	36.0	36.7	43.1	54.0	62.1	70.4	68.3	61.5	53.6	46.4	41.5	(34)
(35)	Mean Daily Temperature Range	MDBR	6.0	6.3	7.0	7.1	8.3	8.0	8.1	7.6	6.6	5.4	5.0	5.5	(35)	
(36)		5% DB	MCDBR	4.7	4.8	6.7	10.4	12.6	11.0	10.7	8.9	7.7	5.4	4.7	4.6	(36)
(37)			MCWBR	4.9	4.5	5.1	6.6	7.0	6.0	5.6	5.3	5.4	4.8	5.2	4.8	(37)
(38)		5% WB	MCDWB	4.8	4.6	6.0	9.6	10.9	9.4	9.2	7.5	6.8	4.9	4.7	4.5	(38)
(39)			MCWBR	5.2	4.4	4.8	6.4	6.8	6.0	5.4	5.3	5.2	5.0	5.3	4.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.283	0.315	0.335	0.372	0.367	0.363	0.391	0.383	0.353	0.336	0.313	0.269	(40)	
(41)		taud	2.167	2.232	2.296	2.321	2.419	2.473	2.419	2.428	2.496	2.404	2.218	2.101	(41)	
(42)		Ebn at Noon	155	216	252	262	273	276	264	257	247	209	141	111	(42)	
(43)		Edh at Noon	14	23	29	34	33	31	33	30	24	19	13	10	(43)	
(44)	All-Sky Solar Radiation	RadAvg	77	242	727	1258	1764	1865	1760	1387	843	366	99	49	(44)	
(45)		RadStd	11	43	86	132	172	118	148	136	103	49	19	6	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-232	N/A	N/A	N/A

Nomenclature: See separate page